TR30 example 8: high-precision appendix

KISSsoft 2026 / ISO 6336:2019. M33 remains UNACCEPTED. Pair with the complete vendor PDF/RTF and native Z12. Values below are COM readbacks after saving, releasing, reopening and recalculating. CSV retains 17 significant digits.

Quantity	Gear 1	Gear 2
YF	1.25190846608	1.27933008798
YS	1.97878024314	2.03527824455
sigmaF0 (MPa)	291.736747614	316.860306892
sigmaF (MPa)	320.0083675	347.566600243
sigmaFP (MPa)	961.667828216	964.625534855
SF	3.00513338364	2.77536890536
sigmaH (MPa)	1284.59135098	1284.59135098
sigmaHP (MPa)	1541.28765916	1541.28765916
SH	1.19982721197	1.19982721197

Protuberance tool and roughness

Quantity	Gear 1	Gear 2
Pre-cut hfP	1.516	1.516
Final hfP (derived)	1.37305845155	1.37305845155
Tool rho_fP	0.4	0.4
Tool hprP	0.632572977907	0.632572977907
Protuberance pr (mm)	0.267	0.267
Finishing stock q (mm)	0.22	0.22
Flank Rz (um)	2.3	2.3
Root Rz (um)	18	18

Tool angle = 10 degrees; residual undercut = pr - q = 0.047 mm. Tool.type=2 (reference profile), Tool.finishing=false: the root remains as pre-cut while the flank is finished. Flank Rz is the printed 2.3 um; it has not been replaced by 6*Ra=2.28 um.

The printed rack values are redundant: pre-cut hfP=1.516 with q=0.220 mm generates final hfP=1.373058451548, which prints as 1.373. Both primary inputs are preserved; the final value is disclosed as a calculated quantity. The tool-height coefficient is an input-coordinate conversion in KISSsoft, inverted from COM tool-geometry evaluations to obtain pr=0.267 mm; it was not fitted to stress results.

Input preservation and comparison limits

All original table rows are listed in input-audit.csv, including dashes and method selections. Measured span is mapped to tooth-thickness allowance by As=(Wtable-Wnominal)/cos(alpha_n). Both allowance limits are set equal; the four spans read back exactly. The parenthesized x2 values are dependent geometry values and agree at the printed precision. For example 7, KISSsoft width includes the 50.8 mm gap: 215.9 mm overall, 165.1 mm active total, 82.55 mm per helix.

An unresolved representation assumption remains: TR30 gives tip relief Ca=13 um but no relief length or shape. KISSsoft overwrites the derived Ca field unless a native modification is present. These files use symmetric linear relief, C=13 um, LCa=0.5*mn, with profile-deflection compensation still disabled. LCa=1*mn is checked separately for sensitivity of every mapped comparison output. Neither length is attributed to TR30, and strict input closure remains false.

Oil nu40 is entered exactly (25 / 220 mm2/s). Other oil thermal, scuffing and micropitting settings inherited from the vendor seed are ancillary defaults absent from TR30; those extra report sections are not TR30 validation evidence. Surface-condition assumptions and any unavailable software quantities remain explicit; M33 is not accepted.

Rows below show remaining differences against printed reference values using the pre-existing v0.8 display-quantum diagnostic. The strict absolute 0.01 AND relative 0.5% columns are separately preserved in comparison.csv. Neither diagnostic grants acceptance.

Quantity	TR30 printed	KISSsoft	Difference
sigma_F0_1	291	291.7367476	0.73674761
sigma_F_2	347	347.5666002	0.56660024

Full line-by-line comparison (including gearmath v0.8 with its original formulas and material conversion) is in comparison.csv / comparison.json. f_sh0 is not exposed in this extraction; no derived convention was silently substituted. Input checks: {'matched': 112, 'not_applicable': 18, 'derived_matches_printed': 3}. Reopen numeric changes: 0.

COM calculation messages were empty, but opening example 7 in the GUI displayed informational warnings about fZCa selection, high circumferential speed/lubrication, scuffing method range and feasibility of span measurement on gear 2. The original table values were retained. Example 8 required an initial second calculation without changing inputs; final and reopened calculations returned true. See run.json for each attempt.